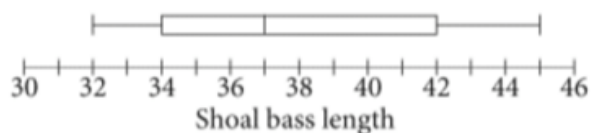


SAT UZBEKISTAN

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1.



A marine biologist is studying a habitat that contains fish called shoal bass. The box plot above summarizes the lengths, in centimeters (cm), of a sample of 65 shoal bass from that habitat. What is the range, in cm, of the lengths of the shoal bass in the sample?

- A) 8
- B) 13
- C) 28
- D) 37

2.

For which of the following sets of 9 data points is the median less than the mean?



3.

Number of pages	Frequency
46	1
47	2
48	4
49	5
50	5
51	2
52	5
53	4

Paulina has 28 books in her collection. The frequency table summarizes the number of pages in each book. If she buys a new book that has 62 pages, how will this impact the mean and median numbers of pages of the books in her collection?

- A) The mean and median will both increase.
- B) The mean and median will both decrease.
- C) The mean will increase, and the median will remain the same.
- D) The mean will decrease, and the median will remain the same.

4.

There are 10 rare coins in a collection. If the value of each of the 10 coins increases by \$2, which of the following will be true?

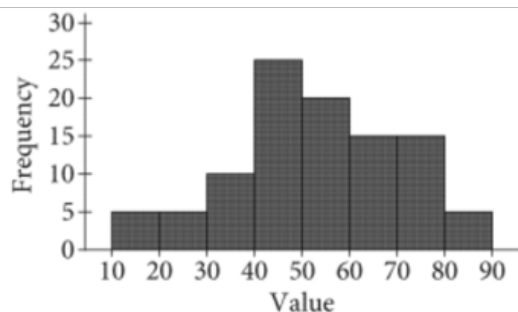
- A) The new mean of the values will be \$2 more than the previous mean, but the standard deviation will remain the same.
- B) The mean of the values will remain the same, but the new standard deviation will be \$2 more than the previous standard deviation.
- C) Both the new mean and standard deviation of the values will be \$2 more than the previous mean and standard deviation.
- D) Neither the mean nor the standard deviation of the values will change.

5.

A data set consists of 50 positive values, and the mean of the data is m . A new data set is made by increasing each value in the original data set by 8. Which of the following represents the mean of the new data set in terms of m ?

- A) $m + 8$
- B) $m + 50$
- C) $8m$
- D) $8m + 50$

6.



The data set summarized by the histogram above contains 100 values. The first bar includes values of more than 10 and at most 20, the second bar includes values of more than 20 and at most 30, and so on. Which range of values contains the median value of the data set?

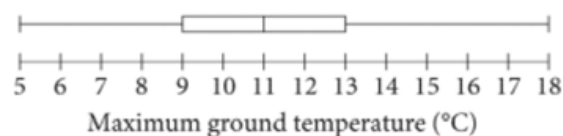
- A) More than 30 and at most 40
- B) More than 40 and at most 50
- C) More than 50 and at most 60
- D) More than 60 and at most 70

7.

A journal article reported the results of a study in which the mean time spent exercising per month was estimated as 10.7 hours, with an associated margin of error of 0.4 hours. A second study of the same population was conducted and included a larger random sample than the random sample from the first study. Based on the change in the sample size, which of the following is the most likely impact on the results?

- A) The mean decreases.
- B) The mean increases.
- C) The margin of error decreases.
- D) The margin of error increases.

8.



The box plot shows the maximum ground temperature, in degrees Celsius ($^{\circ}\text{C}$), each day for 40 solar days on Mars. Which of the following could be the number of solar days for which the maximum ground temperature on Mars was at least 9°C ?

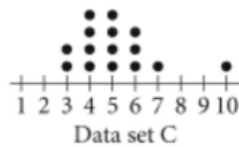
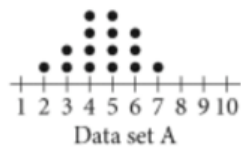
- A) 10
- B) 20
- C) 25
- D) 30

9.

Data set A consists of 20 different values that have a minimum of 22, a maximum of 86, a mean of 54, and a standard deviation of 15. The values 22 and 86 are removed from the set to create data set B , which consists of 18 different values. Which of the following statements is true?

- A) The mean of data set B is greater than 54.
- B) The mean of data set B is less than 54.
- C) The standard deviation of data set B is greater than 15.
- D) The standard deviation of data set B is less than 15.

10. Each of the three data sets represented by the three dot plots has 15 values.



The medians of data sets A, B, and C are a , b , and c , respectively. What is the relationship between a , b , and c ?

- A) $a > b > c$
- B) $b > a > c$
- C) $c > b > a$
- D) $a = b = c$

11.

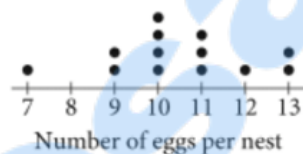


Which statement best compares the means of the two data sets shown?

- A) The mean of data set A is greater than the mean of data set B.
- B) The mean of data set A is equal to the mean of data set B.
- C) The mean of data set A is less than the mean of data set B.
- D) There is not enough information to compare the means.

12.

The dot plot shows the number of eggs in each of 13 wood duck nests.



How many of these nests had exactly 11 eggs?

- A) 1
- B) 2
- C) 3
- D) 4

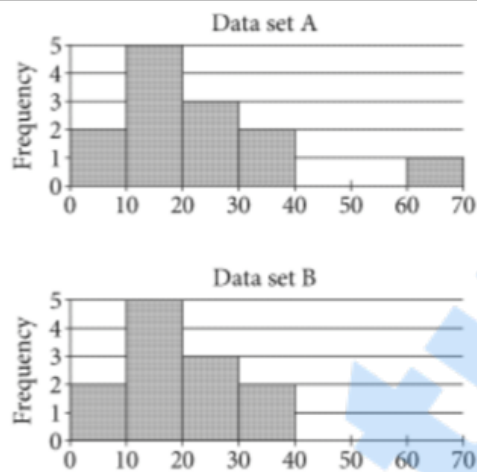
13. The dot plot below shows the number of days it rained in a certain week for 14 cities in the southeastern United States.



Which of the following box plots correctly represents these data?

- A)
- B)
- C)
- D)

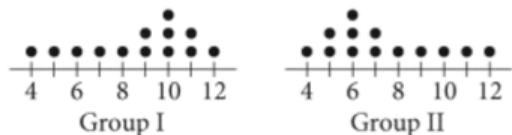
14.



The two histograms show the distribution of data set A and data set B, respectively. Data set B is the result of removing the outlier from data set A. Which of the following statements about the means of data set A and data set B is true?

- A) The means of data set A and B are the same.
- B) The mean of data set A is greater than the mean of data set B.
- C) The mean of data set A is less than the mean of data set B.
- D) No comparison about the means of the data sets can be made.

15.



The dot plots show 13 data values from group I and 13 data values from group II. Which of the following statements best compares the means and standard deviations of the values for each group?

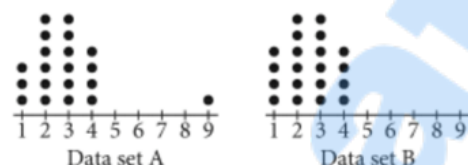
- A) The mean of group I is greater than the mean of group II, and the standard deviation of group I is greater than the standard deviation of group II.
- B) The mean of group II is greater than the mean of group I, and the standard deviation of group II is greater than the standard deviation of group I.
- C) The mean of group I is greater than the mean of group II, but the standard deviations of group I and group II are the same.
- D) The mean of group II is greater than the mean of group I, but the standard deviations of group I and group II are the same.

16.

A sample of 200 bags of flour was selected at random from a production line, and the weights of the bags were analyzed. The mean weight was found to be 79.82 ounces (oz). The margin of error associated with the estimate is 0.24 oz. Based on the estimate and margin of error, which of the following is the most plausible conclusion about the mean weight of all bags of flour produced on the line?

- A) It is less than 79.82 oz.
- B) It is greater than 79.82 oz.
- C) It is less than 79.58 oz or greater than 80.06 oz.
- D) It is between 79.58 oz and 80.06 oz.

17.

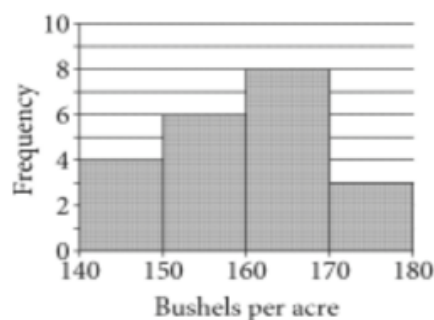


Which statement best compares the means of the two data sets shown?

- A) The mean of data set A is greater than the mean of data set B.
- B) The mean of data set A is equal to the mean of data set B.
- C) The mean of data set A is less than the mean of data set B.
- D) There is not enough information to compare the means.

18.

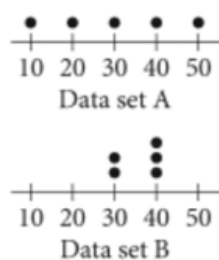
The histogram summarizes the distribution of corn production, in bushels per acre, over a 21-acre farm. The first bar represents the number of acres that produced at least 140 bushels but less than 150 bushels. The second bar represents the number of acres that produced at least 150 bushels but less than 160 bushels, and so on.



The median production, in bushels per acre, of these 21 acres is included in which interval?

- A) At least 140 but less than 150
- B) At least 150 but less than 160
- C) At least 160 but less than 170
- D) At least 170 but less than 180

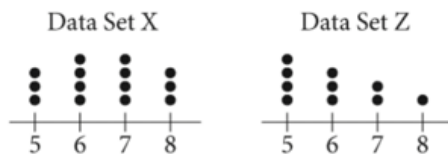
19.



Which of the given data sets has a greater standard deviation?

- A) Data set A
- B) Data set B
- C) The standard deviations of data set A and data set B are equal.
- D) There is not enough information to determine which data set has a greater standard deviation.

20.

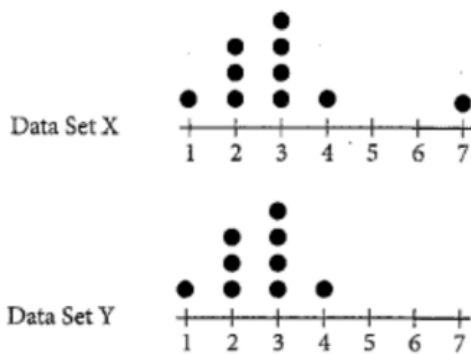


The dot plots above show the distribution of the values of two data sets. Which of the following is(are) true?

- I. The median of the values in data set X is greater than the median of the values in data set Z.
- II. The mean of the values in data set X equals the mean of the values in data set Z.

- A) Neither I nor II
- B) I only
- C) II only
- D) Both I and II

21.

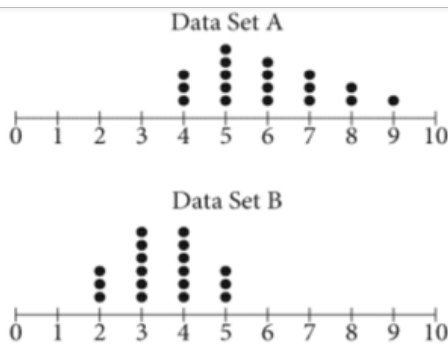


Data set X and data set Y are represented by the dot plots shown. Which of the following is (are) true?

- I. The median of data set X is greater than the median of data set Y.
- II. The mean of data set X is greater than the mean of data set Y.

- A) I only
- B) II only
- C) I and II
- D) Neither I nor II

22.

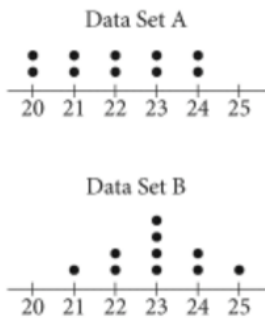


The dot plots above represent the distributions of two data sets. How do the mean and standard deviation of data set A compare with the mean and standard deviation of data set B?

- A) The mean of data set A is greater than that of data set B, and the standard deviation of data set A is less than that of data set B.
- B) The mean of data set A is greater than that of data set B, and the standard deviation of data set A is greater than that of data set B.
- C) The mean of data set A is less than that of data set B, and the standard deviation of data set A is less than that of data set B.
- D) The mean of data set A is less than that of data set B, and the standard deviation of data set A is greater than that of data set B.

23.

Each dot plot represents a data set.



Which of the following statements best compares the means and the standard deviations of the two data sets?

- A) The mean of data set A is less than the mean of data set B, and the standard deviation of data set A is greater than the standard deviation of data set B.
- B) The mean of data set A is greater than the mean of data set B, and the standard deviation of data set A is less than the standard deviation of data set B.
- C) The mean of data set A is less than the mean of data set B, and the standard deviation of data set A is less than the standard deviation of data set B.
- D) The mean of data set A is greater than the mean of data set B, and the standard deviation of data set A is greater than the standard deviation of data set B.

24.

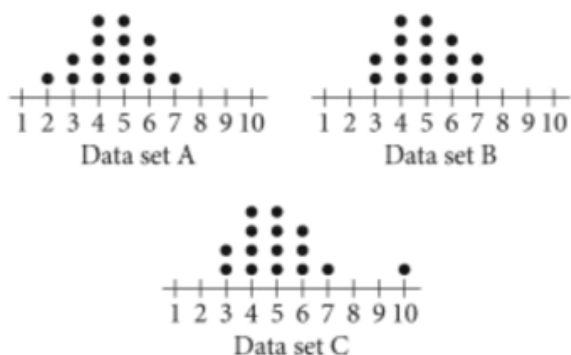
52, 58, 12, 62, 62, 70, 66, 68, 53, 67

The list above shows the number of points a basketball team scored in each of 10 games. During the 10 games, the team scored a total of 570 points. If the outlier is removed from this data set, how will the mean and median change?

- A) Both the mean and the median will increase.
- B) Both the mean and the median will decrease.
- C) The mean will increase and the median will not change.
- D) The mean will decrease and the median will not change.

25.

Each of the three data sets represented by the three dot plots has 15 values.



The medians of data sets A, B, and C are a , b , and c , respectively. What is the relationship between a , b , and c ?

- A) $a > b > c$
- B) $b > a > c$
- C) $c > b > a$
- D) $a = b = c$

26.

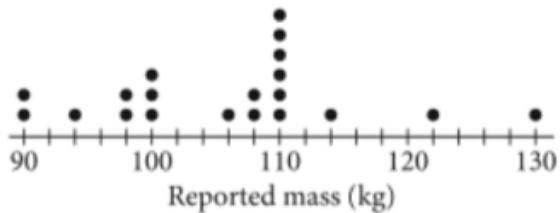
Data set X	7	7	8	8	9	10	10	11	11
Data set Y	1	1	2	2	3	4	4	5	5

The standard deviation of data set X is q , and the standard deviation of data set Y is s . Which of the following statements about the standard deviation of the data sets is true?

- A) $q > s$
- B) $q = s$
- C) $q < s$
- D) The relationship between q and s cannot be determined.

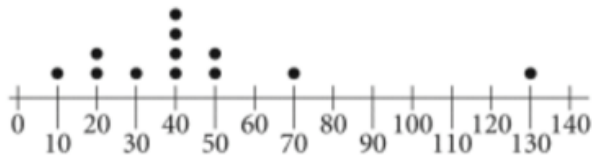
27.

The dot plot shows the reported mass, in kilograms (kg), of 20 ostriches.

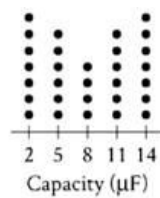


How many of the ostriches have a reported mass of 110 kg?

28.



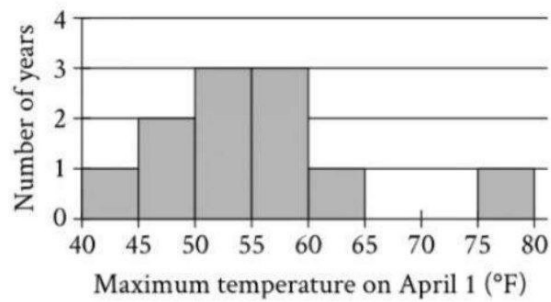
The dot plot above shows the weights, in pounds, of all 12 dogs in a dog park. When one dog leaves the park, the mean weight of the remaining dogs becomes less than the mean weight of the initial 12 dogs in the park. Of the weights of dogs in the dot plot above, what is a possible weight, in pounds, for the dog that left the park?



The dot plot shows the distribution of capacity for a set of capacitors, set A, which a researcher used for a certain experiment. For another experiment, the researcher used a different set of capacitors, set B. Set B has the same number of capacitors as set A but the capacity of each capacitor of set B is 17 microfarads (μF) greater than the capacity of each respective capacitor in set A. Which of the following is true about the capacities of the capacitors in set B?

- (A) The mean capacity is 8 μF , and the range of capacities is 12 μF .
- (B) The mean capacity is 8 μF , and the range of capacities is 29 μF .
- (C) The mean capacity is 25 μF , and the range of capacities is 12 μF .
- (D) The mean capacity is 25 μF , and the range of capacities is 29 μF .

30 The maximum temperature on April 1, in degrees Fahrenheit ($^{\circ}\text{F}$), was recorded each year at a certain weather station for 11 years. The histogram summarizes the recorded data set.



The temperature of 78.2°F is removed from this data set to create a new data set of 10 temperatures. Which of the following statements must be true?

- I. The mean of the new data set is less than the mean of the original data set.
- II. The median of the new data set is less than the median of the original data set.

(A) I only

(B) II only

(C) I and II